

Long Answer Questions [5 Marks each]

1. Discuss the importance of social media platforms in facilitating communication and collaboration between police departments and residents. Provide examples to support your answer.
2. What is Cyber Bullying? What are the types of Cyberbullying? Explain any 3.
3. Discuss the significance of privacy policies in online social media platforms. How do these policies contribute to user trust and data protection? (3 Marks)
Provide examples of specific privacy policies discussed in the lectures to support your answer. (2 Marks)
4. Mention 3 Westin's privacy index. Give one persona for each.
5. How can we detect a voter privacy leak from a tweet to inform the user and prevent it from happening in the future? Explain with an example.

Options :

6406532564116. ✓ I have written answers on the answer sheets

6406532564117. ✖ Not applicable

LLM

Section Id :	64065353062
Section Number :	9
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0

Sub-Section Number : 1
Sub-Section Id : 640653111642
Question Shuffling Allowed : No
Is Section Default? : null

Question Number : 133 Question Id : 640653766126 Question Type : MCQ Display Question Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0
Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INTRODUCTION TO LARGE LANGUAGE MODELS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

- Options :**
- 6406532564118.  YES
 - 6406532564119.  NO

Sub-Section Number : 2
Sub-Section Id : 640653111643
Question Shuffling Allowed : No
Is Section Default? : null

Question Id : 640653766127 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (134 to 137)
Question Label : Comprehension

Three teams, namely A, B, and C, decided to use a GPT model with two decoder layers with the following configuration

- length of context window (T) = 1024
- number of heads $n_h = 8$
- $d_{model} = 512$
- $d_{ff} = 4 * d_{model}$
- $d_q = d_k = d_v = \frac{d_{model}}{n_h} = \frac{512}{8} = 64$
- The weights of the embedding layer and the output layer are shared (tied)

2 dec.

They train the model on the pre-training dataset containing one million words after applying normalization and pre-tokenization(using white space as a delimiter). Moreover, the dataset has no duplicate sentences and no upper-case letters or words. Suppose the team prefers to use the BPE tokenizer to build the vocabulary and subsequently use that to train the model. The base vocabulary contains lowercase alphabets (a to z), digits (0 to 9) and special tokens [unk],[go] and [end].

- Team A uses base vocabulary
- Team B takes the vocabulary from team A and does 500 merges
- Team C takes the vocabulary from the team B and does additional 500 merges

39

539

26
10
3

39

Based on the above data answer the given subquestions.

Sub questions

Question Number : 134 Question Id : 640653766128 Question Type : SA Display Question
Number : Yes Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction
Time : 0

Correct Marks : 3

Question Label : Short Answer Question

What is the size of the vocabulary built by the Team C?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1039

Question Number : 135 Question Id : 640653766129 Question Type : MSQ Display Question

Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time :

N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Choose the correct statements about the number of parameters in the model (excluding the embedding and output layer parameters)

Options :

6406532564121. ✖ Team A model has less number of parameters than the team B

6406532564122. ✖ Team B model has less number of parameters than the team C

6406532564123. ✖ Team C model has more number of parameters than the team A

6406532564124. ✔ All models have the same number of parameters

Question Number : 136 Question Id : 640653766130 Question Type : MSQ Display Question

Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time :

N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Assume that the word "acrophobia" is not present in the vocabulary, then which of the following tokenizer(s) is(are) guaranteed to tokenize this word into sub-words units (i.e., it does not return [unk] token)

Options :

6406532564125. ✔ The tokenizer used by the team A

6406532564126. ✔ the tokenizer used by the team B

6406532564127. ✓ the tokenizer used by the team C

6406532564128. ✗ None of the given options

Question Number : 137 Question Id : 640653766131 Question Type : MCQ Display Question Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

1×10^6

Assume that team A and team B pre-trained the model for 1 million steps. Suppose they want to fine-tune the model for sentiment classification tasks (where the dataset contains sentences, digits and emojis). Choose the correct statement

Options :

6406532564129. ✓ Team A can use the tokenizer from Team B and fine-tune the model and vice versa

6406532564130. ✗ We can not interchange both teams' tokenizers while fine-tuning.

Sub-Section Number : 3

Sub-Section Id : 640653111644

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 138 Question Id : 640653766132 Question Type : MSQ Display Question Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose we have two decoder-only models, namely Model A and Model B. Assume we have two datasets, one for unsupervised pre-training (Web-Text) and the other for supervised training (text summarization). Model A was trained directly on the training set of the supervised dataset. Model B was trained on the pre-training dataset using the CLM objective. Suppose we want to test the performance of both models on the test set of summarization task. Then choose the correct

statements

Options :

6406532564131. ✓ Model A can directly be used on the test set

6406532564132. ✓ It is possible to use Model B directly on the test set without fine-tuning on the training set

6406532564133. ✗ Model B **should be** fine-tuned first on the training set and then use the test set.

Question Number : 139 Question Id : 640653766137 Question Type : MSQ Display Question Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

The strikethrough words in the passage given below denote the words to be dropped from the original sentence. "At the heart of science is an essential balance between two seemingly ~~contradictory~~ attitudes — an openness to new ideas, no matter how bizarre or ~~counterintuitive~~ they may be, and the most ruthless skeptical scrutiny of all ideas, old and new. This is ~~how~~ deep truths are winnowed from deep nonsense"

Which of the following represents the correct input sequence, with sentinel tokens, to the baseline model that uses the pre-training denoising objective? [z] represents the end of the sentinel token in a sentence. The characters inside the square brackets are the sentinel tokens

Options :

6406532564145. ✗ [v] of science is [w] contradictory attitudes [x] counterintuitive they may be [y] how deep truths are winnowed [z]

6406532564146. ✗ [a] of science is [b] contradictory attitudes [c] counterintuitive they may be [d] how deep truths are winnowed [z]

6406532564147. ✓ [v] of science is [w] contradictory attitudes [x] counterintuitive they may be [y] how deep truths are winnowed

6406532564148. ✓ [a] of science is [b] contradictory attitudes [c] counterintuitive they may be [d]
how deep truths are winnowed

6406532564149. ✖ [w] contradictory attitudes [v] of science is [x] counterintuitive they may be [y]
how deep truths are winnowed [z]

6406532564150. ✖ [v] of science is [v] contradictory attitudes [x] counterintuitive they may be [x]
how deep truths are winnowed [y]

Sub-Section Number : 4
Sub-Section Id : 640653111645
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 140 Question Id : 640653766133 Question Type : MSQ Display Question
Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time :
N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following models uses a form of denoising objective for pre-training?

Options :

6406532564134. ✓ BERT

6406532564135. ✖ GPT-1 and GPT-2

6406532564136. ✓ BART

6406532564137. ✓ T5

Sub-Section Number : 5
Sub-Section Id : 640653111646
Question Shuffling Allowed : No
Is Section Default? : null

Question Id : 640653766134 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (141 to 142)

Question Label : Comprehension

Consider training an encoder-decoder model (say, T5 baseline) on a pre-training dataset that contains 1 Billion tokens. Let the batch size be 64×128 , that is 2^{13} tokens. Suppose the model is trained for 2^{15} steps by varying the size of the data (all other aspects are fixed)

A. Train the model only on the first one million tokens, referred to as Model A

B. Train the model on the entire dataset, referred to as Model B

Based on the above data answer the given subquestions.

Sub questions

Question Number : 141 Question Id : 640653766135 Question Type : MSQ Display Question

Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time :

N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements is (are) correct under this setup?

Options :

6406532564138. ✓ Training loss of Model A will be less than training loss of Model B

6406532564139. ✗ Training loss of Model B will be less than training loss of Model A

6406532564140. ✓ Test loss of Model B will likely be less than test loss of Model A

6406532564141. ✗ Test loss of Model A will likely be less than test loss of Model B

Question Number : 142 Question Id : 640653766136 Question Type : MSQ Display Question

Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time :

N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following modifications will improve the performance of both models?

Options :

6406532564142. ✓ Increasing the training steps helps Model B more than Model A in terms of generalization performance

6406532564143. ✗ Increasing the training steps helps both Model A and Model B in terms of generalization performance

6406532564144. ✗ None of the given modifications helps the model to improve their generalization performance

Sub-Section Number :

6

Sub-Section Id :

640653111647

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 143 Question Id : 640653766138 Question Type : MSQ Display Question

Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time :

N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a summarization task. Assume the prefix is "Summarize the following:" and the input sentence is "There are a lot of videos on YouTube etc. that give the appearance of education, but if you look closely they are really just entertainment. This is very convenient for everyone involved: the people watching enjoy thinking they are learning (but they are just having fun)" and the model is supposed to summarize it. Then which of the following architectures **could be used** for this task

Options :

6406532564151. ✓ A decoder-only model with a causal mask

6406532564152. ✓ A decoder-only model with a prefix mask

6406532564153. ✗ An encoder-only model with a fully visible mask

~~6406532564154.~~ ✓ Encoder-Decoder model with fully visible mask in the encoder and causal mask in the decoder

Sub-Section Number : 7
Sub-Section Id : 640653111648
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 144 Question Id : 640653766139 Question Type : MSQ Display Question
Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time :
N.A Minimum Instruction Time : 0
Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose we are given a Model A with N layers and the dataset with D tokens. Choose the correct statements according to the scaling law.

Options :

~~6406532564155.~~ ✓ Increasing the model size requires a proportionate increase in the dataset size to reduce the test loss

6406532564156. ✖ Increasing the model size requires a proportionate increase in the dataset size to reduce the training loss

6406532564157. ✖ Increasing the model size requires us to double the dataset size to reduce the test loss

6406532564158. ✖ Increasing the model size requires us to double the dataset size to reduce the train loss

Sub-Section Number : 8
Sub-Section Id : 640653111649
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 145 Question Id : 640653766140 Question Type : MCQ Display Question

Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following components in the data pre-processing pipeline removes pages that contain bad words?

Options :

6406532564159. ✖ Language Identification

6406532564160. ✖ Exact Deduplication

6406532564161. ✖ Fuzzy Deduplication

6406532564162. ✖ ML classifiers for quality filtering

6406532564163. ✔ Simple heuristics to detect toxic contents

Question Number : 146 Question Id : 640653766141 Question Type : MCQ Display Question

Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Suppose a team wants to build a dataset for a low-resource language. They took all Wikipedia dumps for that language and found that the dataset size was insufficient. Then the statement that using years of Common Crawl Snapshots would be helpful to increase the size of the dataset is

Options :

6406532564164. ✔ TRUE

6406532564165. ✖ FALSE

Sub-Section Number :

9

Sub-Section Id :

640653111650

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 147 Question Id : 640653766142 Question Type : MCQ Display Question Number : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Correct Marks : 2

Question Label : Multiple Choice Question

The statement that the presence of significant duplicate contents in the pre-training dataset degrades the performance of Large Language Models is

Options :

6406532564166. ✓ TRUE

6406532564167. ✗ FALSE

DL(CV)

Section Id :	64065353063
Section Number :	10
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	23
Number of Questions to be attempted :	23
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653111651